Fingertip Trajectory Control Robotic Arm

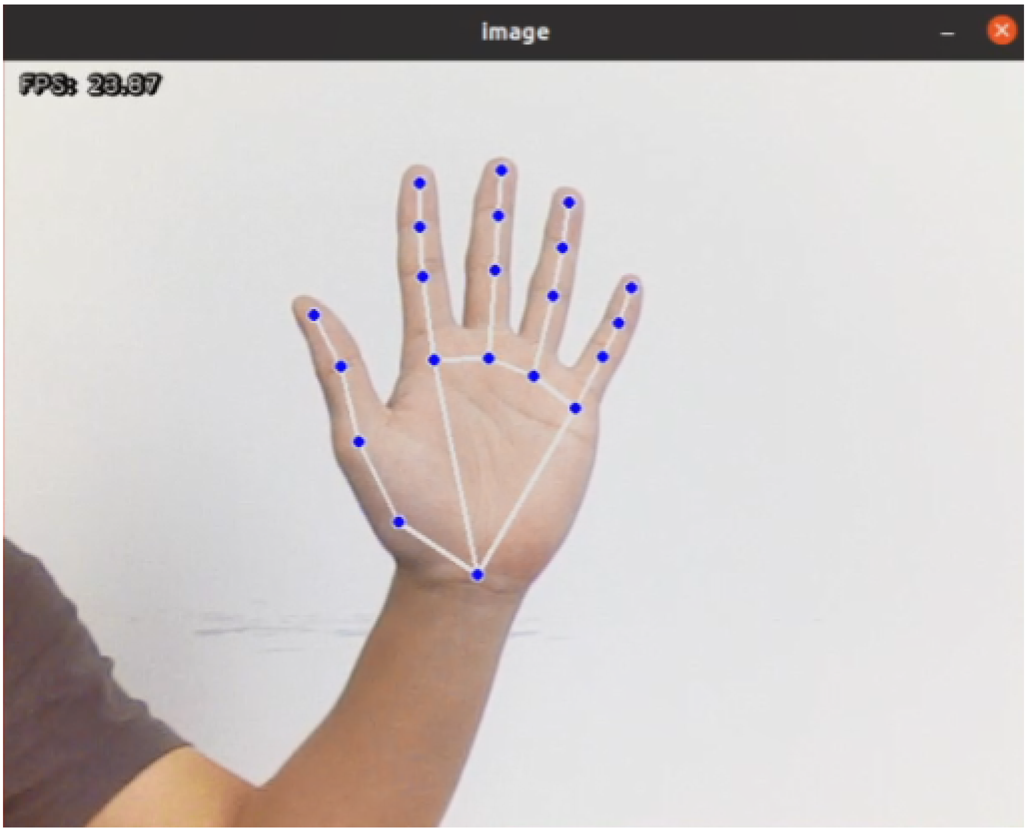
1. Introduction

The fingertip trajectory control robotic arm function is based on fingertip trajectory recognition, with the added capability of different trajectories controlling different robotic arm actions.

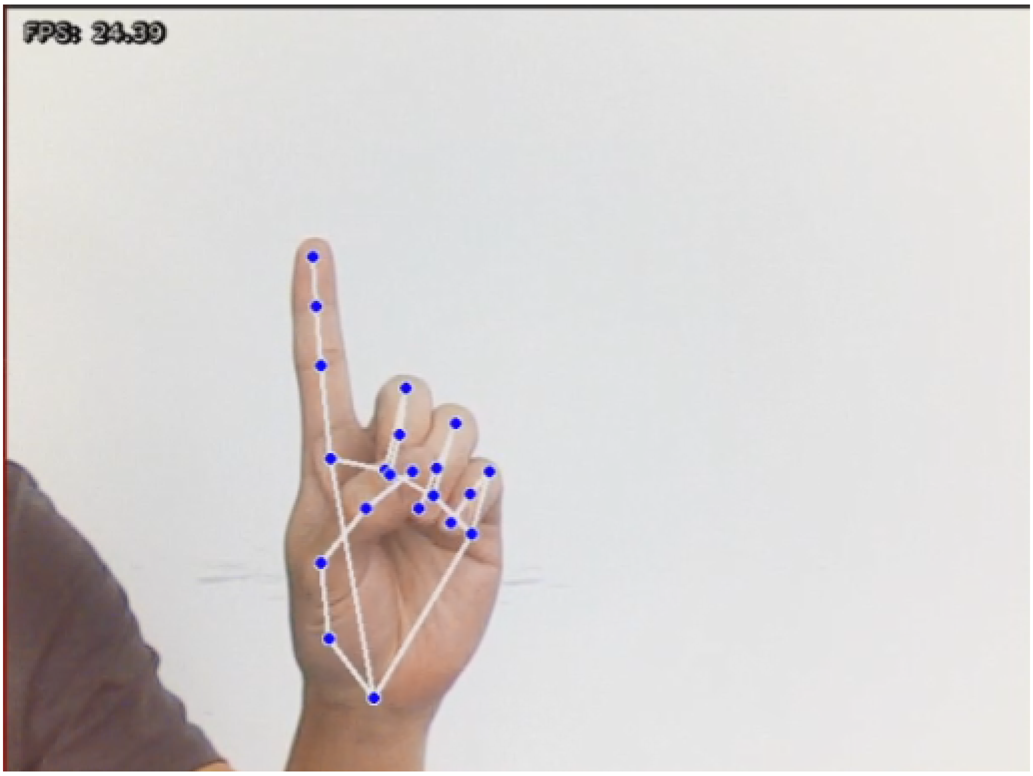
2. Startup

2.1. Program Description

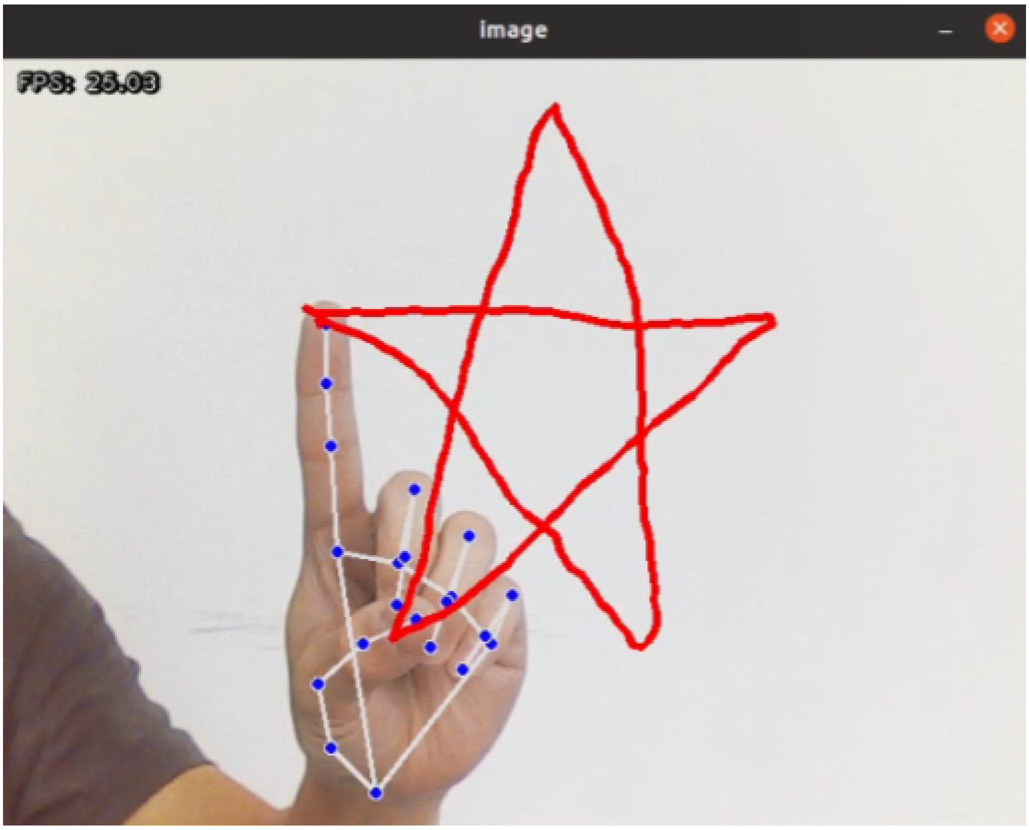
After the program starts, the camera captures the image. Place your palm flat in the camera frame, open your fingers, with your palm facing the camera, similar to the number 5 gesture. The image will draw all the joints on the palm. Adjust the palm position to be roughly in the upper middle part of the screen.



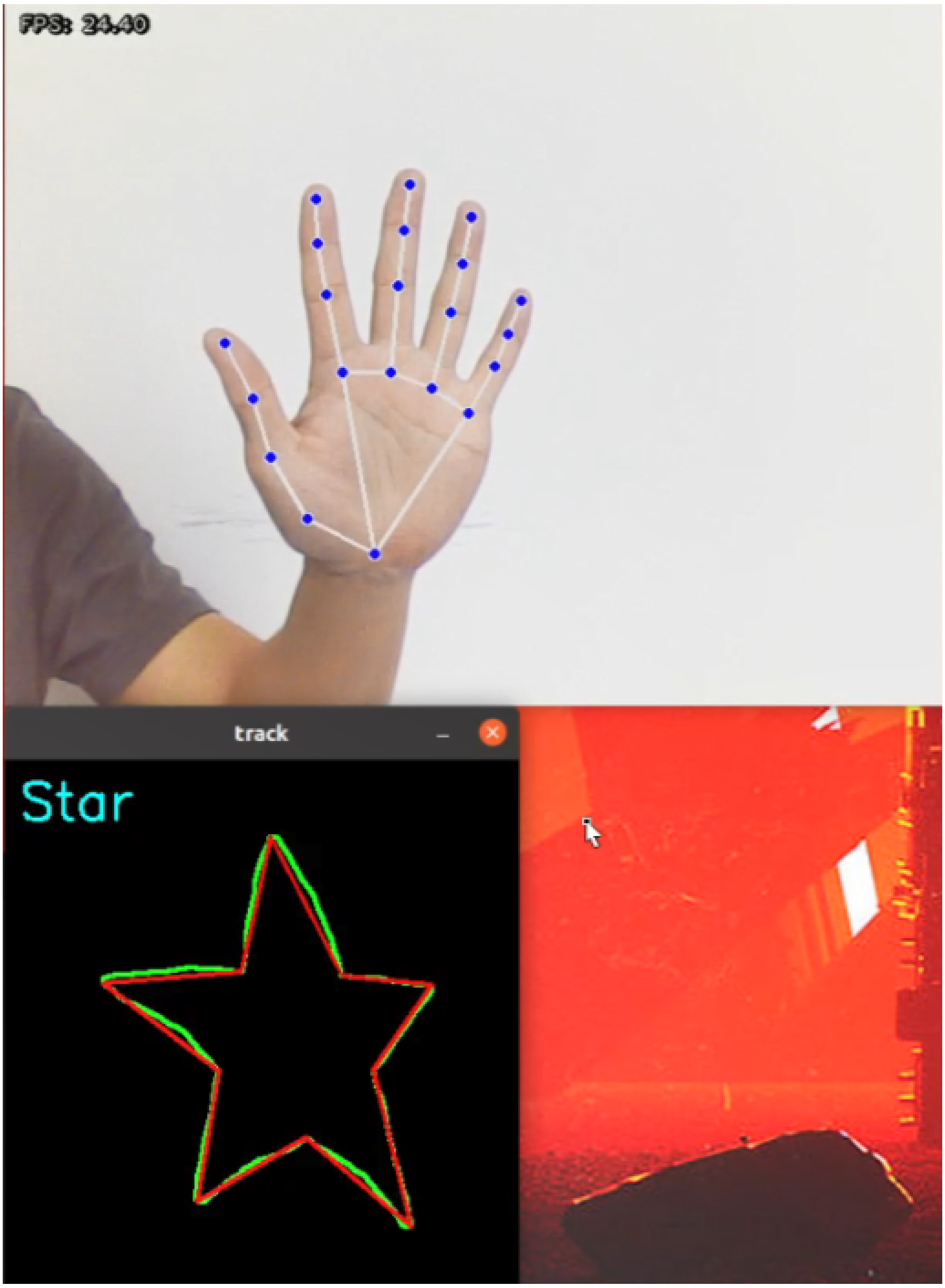
Now keep the index finger unchanged and retract the other fingers, similar to the number 1 gesture.



While keeping the gesture 1 posture unchanged, move your finger position. Red lines will appear on the screen, drawing the movement path of the index finger.



When the drawing is complete, open all fingers, similar to the number 5 gesture. The drawn shape will be generated below.



Note: The drawn shape needs to be closed, otherwise some content may be missing.

Currently, there are four recognizable trajectory shapes: triangle, square, circle, and five-pointed star.

When the camera recognizes different trajectory shapes, it will control the robotic arm to perform corresponding actions.

2.2. Program Startup

- Enter the following command to start the program

```
ros2 run dofbot_mediapipe 16_FingerAction
```

Press Ctrl+c in the terminal to exit the program.

3. Source Code

Code path:

```
/home/yahboom/dofbot_ws/src/dofbot_mediapipe/dofbot_mediapipe/16_FingerAction.py
```

```
#!/usr/bin/env python3
# coding: utf8
import os
import enum
import cv2
import time
import numpy as np
import mediapipe as mp
import rclpy
from rclpy.node import Node
import queue
from sensor_msgs.msg import Image
from dofbot_utils.fps import FPS
import gc
from dofbot_utils.vutils import distance, vector_2d_angle, get_area_max_contour
import threading
from Arm_Lib import Arm_Device
from dofbot_utils.robot_controller import Robot_Controller
from cv_bridge import CvBridge  # Importing CvBridge


def get_hand_landmarks(img, landmarks):
    """
    Convert landmarks from mediapipe normalized output to pixel coordinates
    :param img: Image corresponding to pixel coordinates
    :param landmarks: Normalized keypoints
    :return:
    """
    h, w, _ = img.shape
    landmarks = [(lm.x * w, lm.y * h) for lm in landmarks]
    return np.array(landmarks)


def hand_angle(landmarks):
    """
    Calculate the bending angle of each finger
    :param landmarks: Hand keypoints
    :return: Angle of each finger
    """
    angle_list = []
    # thumb
    angle_ = vector_2d_angle(landmarks[3] - landmarks[4], landmarks[0] -
landmarks[2])
    angle_list.append(angle_)
    # index
    angle_ = vector_2d_angle(landmarks[0] - landmarks[6], landmarks[7] -
landmarks[8])
    angle_list.append(angle_)
    # middle
```

```
    angle_ = vector_2d_angle(landmarks[0] - landmarks[10], landmarks[11] -
landmarks[12])
    angle_list.append(angle_)
    # ring
    angle_ = vector_2d_angle(landmarks[0] - landmarks[14], landmarks[15] -
landmarks[16])
    angle_list.append(angle_)
    # pink
    angle_ = vector_2d_angle(landmarks[0] - landmarks[18], landmarks[19] -
landmarks[20])
    angle_list.append(angle_)
    angle_list = [abs(a) for a in angle_list]
    return angle_list


def h_gesture(angle_list):
    """
    Determine the finger gesture through 2D features
    :param angle_list: Bending angle of each finger
    :return : Gesture name string
    """
    thr_angle, thr_angle_thumb, thr_angle_s = 65.0, 53.0, 49.0
    if (angle_list[0] < thr_angle_s) and (angle_list[1] < thr_angle_s) and
(angle_list[2] < thr_angle_s) and (
            angle_list[3] < thr_angle_s) and (angle_list[4] < thr_angle_s):
        gesture_str = "five"
    elif (angle_list[0] > 5) and (angle_list[1] < thr_angle_s) and
(angle_list[2] > thr_angle) and (
            angle_list[3] > thr_angle) and (angle_list[4] > thr_angle):
        gesture_str = "one"
    else:
        gesture_str = "none"
    return gesture_str


class State(enum.Enum):
    NULL = 0
    TRACKING = 1
    RUNNING = 2


def draw_points(img, points, tickness=4, color=(255, 0, 0)):
    """
    Draw connecting lines of recorded points on the image
    """
    points = np.array(points).astype(dtype=np.int32)
    if len(points) > 2:
        for i, p in enumerate(points):
            if i + 1 >= len(points):
                break
            cv2.line(img, tuple(p), tuple(points[i + 1]), color, tickness)


def get_track_img(points):
    """
    Generate a black background white line trajectory image from recorded points
    """
    points = np.array(points).astype(dtype=np.int32)
```

```python
    x_min, y_min = np.min(points, axis=0).tolist()
    x_max, y_max = np.max(points, axis=0).tolist()
    track_img = np.full([y_max - y_min + 100, x_max - x_min + 100, 1], 0,
dtype=np.uint8)
    points = points - [x_min, y_min]
    points = points + [50, 50]
    draw_points(track_img, points, 1, (255, 255, 255))
    return track_img


class FingerActionNode(Node):
    def __init__(self):
        super().__init__('finger_action')
        self.drawing = mp.solutions.drawing_utils
        self.timer = time.time()
        self.move_state = False

        self.hand_detector = mp.solutions.hands.Hands(
            static_image_mode=False,
            max_num_hands=1,
            min_tracking_confidence=0.05,
            min_detection_confidence=0.6
        )

        self.fps = FPS()  # fps calculator
        self.state = State.NULL
        self.points = []
        self.start_count = 0
        self.no_finger_timestamp = time.time()

        self.gc_stamp = time.time()
        self.image_queue = queue.Queue(maxsize=1)
        self.bridge = CvBridge()

        # Initialize video capture device
        self.cap = cv2.VideoCapture(0, cv2.CAP_V4L2)
        self.cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
        self.cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)
        if not self.cap.isOpened():
            self.get_logger().error("Error: Could not open video device.")
            rclpy.shutdown()

        self.Arm = Arm_Device()
        self.robot = Robot_Controller()
        self.robot.move_init_pose()

    def image_proc(self):
        ret, frame = self.cap.read()
        if not ret:
            self.get_logger().error("Error: Could not read frame from video
device.")
            return

        rgb_image = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
        rgb_image = cv2.flip(rgb_image, 1)  # horizontal flip
        result_image = np.copy(rgb_image)
        result_call = None
        if self.timer <= time.time() and self.state == State.RUNNING:
```

```
            self.state = State.NULL
        try:
            results = self.hand_detector.process(rgb_image) if self.state !=
State.RUNNING else None
            if results is not None and results.multi_hand_landmarks:
                gesture = "none"
                index_finger_tip = [0, 0]
                self.no_finger_timestamp = time.time()  # record current time
for timeout handling
                for hand_landmarks in results.multi_hand_landmarks:
                    self.drawing.draw_landmarks(
                        result_image,
                        hand_landmarks,
                        mp.solutions.hands.HAND_CONNECTIONS)
                    landmarks = get_hand_landmarks(rgb_image,
hand_landmarks.landmark)
                    angle_list = (hand_angle(landmarks))
                    gesture = (h_gesture(angle_list))
                    index_finger_tip = landmarks[8].tolist()

                if self.state == State.NULL:
                    if gesture == "one":  # detected only index finger extended,
other fingers closed
                        self.start_count += 1
                        if self.start_count > 20:
                            self.state = State.TRACKING
                            self.points = []
                    else:
                        self.start_count = 0

                elif self.state == State.TRACKING:
                    if gesture == "five":  # extend five fingers to end drawing
                        self.state = State.NULL

                        # generate black and white trajectory image
                        track_img = get_track_img(self.points)
                        contours = cv2.findContours(track_img,
cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_NONE)[-2]
                        contour = get_area_max_contour(contours, 300)
                        contour = contour[0]
                        # recognize drawn shape from trajectory image
                        # cv2.fillPoly draws and fills polygons on the image
                        track_img = cv2.fillPoly(track_img, [contour, ], (255,
255, 255))
                        for _ in range(3):
                            # erosion function
                            track_img = cv2.erode(track_img,
cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5)))
                            # dilation function
                            track_img = cv2.dilate(track_img,
cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5)))
                        contours = cv2.findContours(track_img,
cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_NONE)[-2]
                        contour = get_area_max_contour(contours, 300)
                        contour = contour[0]
                        h, w = track_img.shape[:2]

                        track_img = np.full([h, w, 3], 0, dtype=np.uint8)
```

```
                        track_img = cv2.drawContours(track_img, [contour, ], -1,
(0, 255, 0), 2)
                        # polygon fitting of image contour points
                        approx = cv2.approxPolyDP(contour, 0.026 *
cv2.arcLength(contour, True), True)
                        track_img = cv2.drawContours(track_img, [approx, ], -1,
(0, 0, 255), 2)
                        graph_name = 'unknown'
                        print(len(approx))
                        # determine shape based on the number of vertices in the
contour envelope
                        if len(approx) == 3:
                            graph_name = 'Triangle'
                        if len(approx) == 4 or len(approx) == 5:
                            graph_name = 'Square'
                        if 5 < len(approx) < 10:
                            graph_name = 'Circle'
                        if len(approx) == 10:
                            graph_name = 'Star'
                        cv2.putText(track_img, graph_name, (10, 40),
cv2.FONT_HERSHEY_SIMPLEX, 1.2, (255, 255, 0), 2)
                        cv2.imshow('track', track_img)
                        if not self.move_state:
                            self.move_state = True
                            task = threading.Thread(target=self.arm_move_action,
name="arm_move_action", args=(graph_name,))
                            task.setDaemon(True)
                            task.start()

                    else:
                        if len(self.points) > 0:
                            if distance(self.points[-1], index_finger_tip) > 5:
                                self.points.append(index_finger_tip)
                        else:
                            self.points.append(index_finger_tip)

                    draw_points(result_image, self.points)
                else:
                    pass
            else:
                if self.state == State.TRACKING:
                    if time.time() - self.no_finger_timestamp > 2:
                        self.state = State.NULL
                        self.points = []

        except BaseException as e:
            self.get_logger().error("e = {}".format(e))

        self.fps.update_fps()
        self.fps.show_fps(result_image)
        result_image = cv2.cvtColor(result_image, cv2.COLOR_RGB2BGR)
        cv2.imshow('image', result_image)
        key = cv2.waitKey(1)

        if key == ord(' '):  # press space to clear recorded trajectory
            self.points = []
        if time.time() > self.gc_stamp:
            self.gc_stamp = time.time() + 1
```

```python
            gc.collect()

    def arm_move_triangle(self):
        self.Arm.Arm_serial_servo_write6_array([90, 131, 52, 0, 90, 180], 1500)
        time.sleep(1.5)
        self.Arm.Arm_serial_servo_write6_array([45, 180, 0, 0, 90, 180], 1500)
        time.sleep(2)
        self.Arm.Arm_serial_servo_write6_array([135, 180, 0, 0, 90, 180], 1500)
        time.sleep(2)
        self.Arm.Arm_serial_servo_write6_array([90, 131, 52, 0, 90, 180], 1500)
        time.sleep(1.5)

    def arm_move_square(self):
        self.Arm.Arm_serial_servo_write6_array(self.robot.P_ACTION_4, 1500)
        time.sleep(1.4)
        for i in range(3):
            self.Arm.Arm_serial_servo_write(4, -15, 300)
            time.sleep(0.4)
            self.Arm.Arm_serial_servo_write(4, 20, 300)
            time.sleep(0.4)

    def arm_move_circle(self):
        for i in range(5):
            self.Arm.Arm_serial_servo_write(5, 60, 300)
            time.sleep(0.4)
            self.Arm.Arm_serial_servo_write(5, 120, 300)
            time.sleep(0.4)
        self.Arm.Arm_serial_servo_write(5, 90, 300)
        time.sleep(0.4)

    def arm_move_star(self):
        for i in range(3):
            self.Arm.Arm_serial_servo_write6_array(self.robot.P_ACTION_3, 1200)
            time.sleep(1.2)
            self.Arm.Arm_serial_servo_write6_array(self.robot.P_LOOK_AT, 1000)
            time.sleep(1)

    def arm_move_action(self, name):
        self.Arm.Arm_Buzzer_On(1)
        time.sleep(1)
        if name == 'Triangle':
            self.arm_move_triangle()
        elif name == 'Square':
            self.arm_move_square()
        elif name == 'Circle':
            self.arm_move_circle()
        elif name == 'Star':
            self.arm_move_star()
        self.robot.move_init_pose()
        time.sleep(1.5)
        self.move_state = False


def main(args=None):
    rclpy.init(args=args)
    finger_node = FingerActionNode()
    try:
        while rclpy.ok():
```

```python
            finger_node.image_proc()
    except KeyboardInterrupt:
        pass
    finally:
        finger_node.cap.release()
        cv2.destroyAllWindows()
        rclpy.shutdown()


if __name__ == "__main__":
    main()
```